

Investigation of Cross-Validation in Diagnosis of Shared Frailty Models with Z-Residuals

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Section 1

Introduction

- No statistical model comes without any assumptions.
- Diagnostic methods are useful in all types of regression models to investigate the validity of those assumptions and identify ways in which they might be violated.
- Residuals play a central role in diagnosing regression models.
- For diagnosing survival models, Cox-Snell, Martingale and deviance residuals have been widely used, but each tool has its limitations.

Introduction

- One drawback to the **Cox-Snell residuals** is that they don't provide much insight into why the model's assumptions are violated.

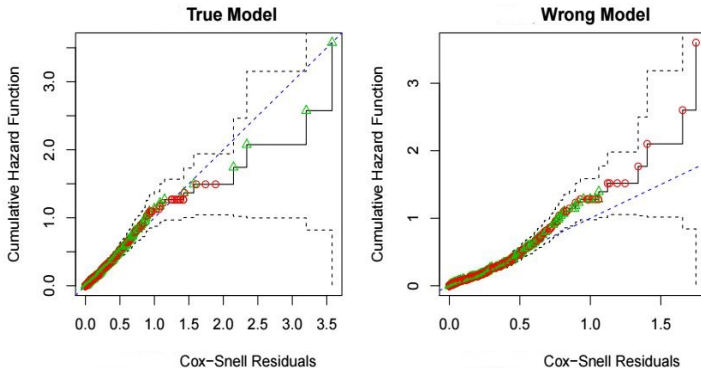


Figure 1: True model: Weibull AFT; Wrong model: log-normal AFT with 80% censoring rate.

Introduction

- **Martingale residuals** are very useful and can be used for many of the usual purposes that we use residuals for in other models. However, the primary drawback to the martingale residual is its clear asymmetry (its upper bound is 1, but it has no lower bound).
- **Deviance residuals** are a transform of martingale residuals. They are symmetric, but there is still a lack of reference distribution.
- It would be more appealing if a residual diagnosis tool can be designed that is **approximately normally distributed under the true model** so that Shapiro-Wilk (SW) normality test can be then used as an overall model GOF test.

Introduction

- A new model diagnostics tool for censored regression via **randomized survival probabilities** by Li, L. et.al (2021)¹.
- The **key idea** is to replace the survival probability of a censored failure time with a uniform random number between 0 and survival probability of the censored time.
- Our simulation studies and real data analysis showed the good power of the proposed method for diagnosing survival models.
- **Goal of the current study was two-fold:**
 - Extend this new model diagnosis tool (z residuals) to diagnose **shared frailty models**.
 - Demonstrate **Cross validation** to address the conservatism of the residual diagnosis.

¹Li, L, Wu, T, Feng, C. Model diagnostics for censored regression via randomized survival probabilities. Statistics in Medicine. 2021; 40: 1482– 1497.

Section 2

Definition of Z residual

Shared frailty models

Suppose there are g groups of individuals with n_i individuals in the i th group, $i = 1, 2, \dots, g$. For the shared frailty models, the hazard of death at time t for the j th individual, $j = 1, 2, \dots, n_i$, in the i th group, is then

$$h_{ij}(t) = z_i \exp(\beta' x_{ij}) h_0(t) \quad (1)$$

where x_{ij} is a vector of values of p explanatory variables for the j th individual in the i th group, β is the vector of their coefficients, $h_0(t)$ is the baseline hazard function, and the z_i are frailty effects that are common for all n_i individuals within the i th group.

Definition of Z residual

The randomized survival probability (RSP)¹ for t_{ij} is defined as:

$$S_{ij}^R(t_{ij}, d_{ij}, U_{ij}) = \begin{cases} S_{ij}(t_{ij}), & \text{if } t_{ij} \text{ is uncensored, } d_{ij} = 1, \\ U_{ij} S_{ij}(t_{ij}), & \text{if } t_{ij} \text{ is censored, } d_{ij} = 0, \end{cases} \quad (2)$$

where U_{ij} is a uniform random number on $(0, 1)$, and $S_{ij}(\cdot)$ is the postulated survival function for T given x_{ij} . $S_{ij}^R(t_{ij}, d_{ij}, U_{ij})$ is a random number between 0 and $S_{ij}(t_{ij})$ when t_{ij} is censored.

¹Li, L, Wu, T, Feng, C. Model diagnostics for censored regression via randomized survival probabilities. Statistics in Medicine. 2021; 40: 1482– 1497.

Definition of Z residual

- It is preferred to transfer RSP with the normal quantile¹ :

$$r_{ij}^Z(t_{ij}, d_{ij}, U_{ij}) = \Phi^{-1}(S_{ij}^R(t_{ij}, d_{ij}, U_{ij})). \quad (3)$$

where $\Phi()$ is the cumulative distribution function (CDF) of a standard normal distribution. We name the residuals in (3) as Z residuals.

- Cross validation (CV) is used to calculate the Z residuals.
 - Leave one out cross validation (LOOCV)
 - 10-fold cross validation

¹Li, L, Wu, T, Feng, C. Model diagnostics for censored regression via randomized survival probabilities. Statistics in Medicine. 2021; 40: 1482– 1497.

Section 3

Simulation Study

Subsection 1

Simulation Studies and Results for Detection of Non-linear Covariate Effect

Simulation Setup

- Covariates $x_{ij}^{(1)} \sim \text{Uniform}(0, 1)$, $x_{ij}^{(2)} \sim \text{positive Normal}(0, 1)$, $x_{ij}^{(3)} \sim \text{Bern}(0.25)$.
- Frailty $z_i \sim \text{gamma}(2, 0.5)$.
- Censored time $C_{ij} \sim \text{Exp}(\theta)$.
- Event times $T_{ij}^* \sim \text{Weibull baseline hazard function with shape } (\alpha) \text{ parameter is 3 and scale } (\lambda) \text{ parameter is 0.007.}$

$$T_{ij}^* = \{-\log(v)/[\lambda z_i \exp(x_{ij}^{(1)}) + \beta_2 \log(x_{ij}^{(2)}) + 0.5x_{ij}^{(3)}]\}^{(1/\alpha)} \quad (4)$$

with $i = \{1, \dots, g\}$ and $j = \{1, \dots, n_i\}$, where v is simulated from uniform distribution, z_i the frailty term of all subjects in group i , $\beta_2 = -2$ (Strong non-linearity) or -1 (Moderate non-linearity).

- $T_{ij} = \min(T_{ij}^*, C_i)$ is the observed survival time.
- We set up $g=10$ and n_i ($n_i = J$) from 10 to 100;
- Censoring rates: 20%, 50% and 80%.

Z residual for a single dataset

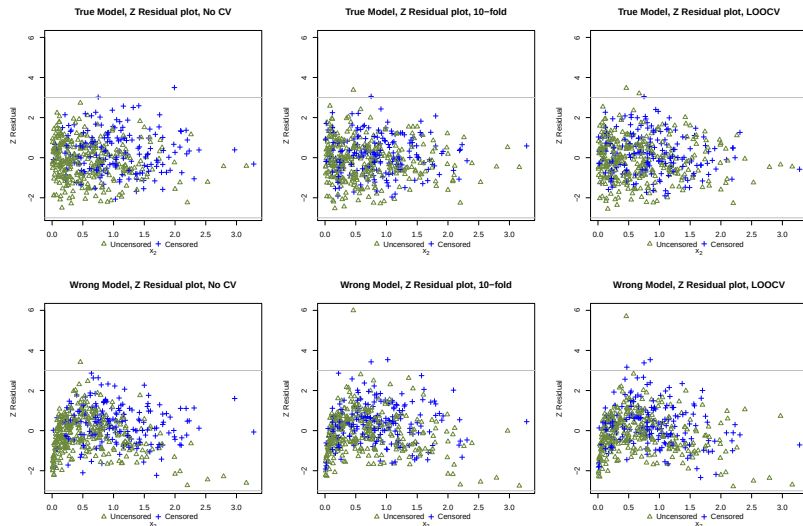
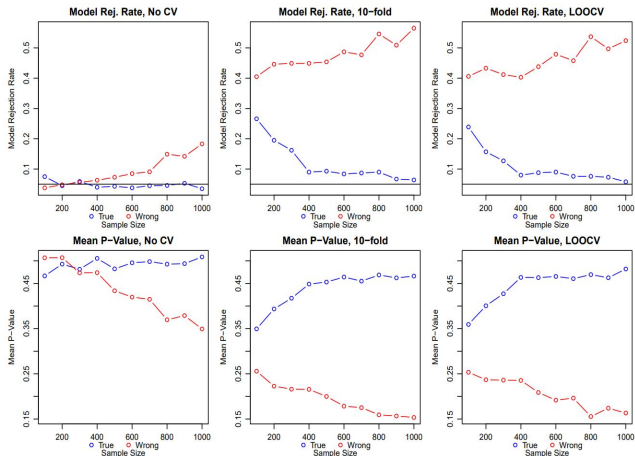
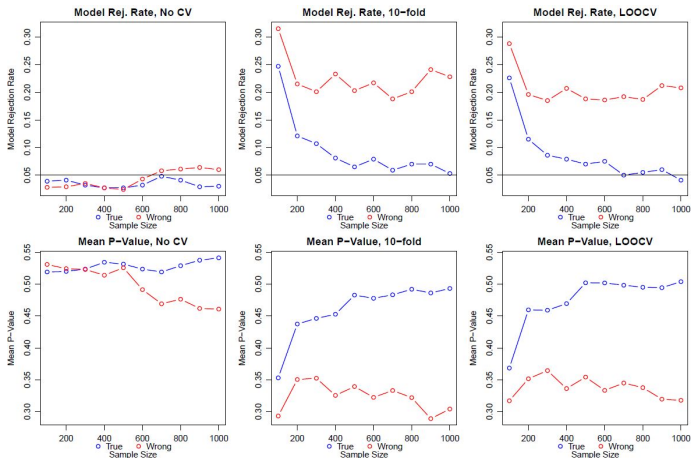


Figure 2: The sample size is 500 (10 clusters of 50 observations), and censoring percentage is 50%.

Strong non-linear association

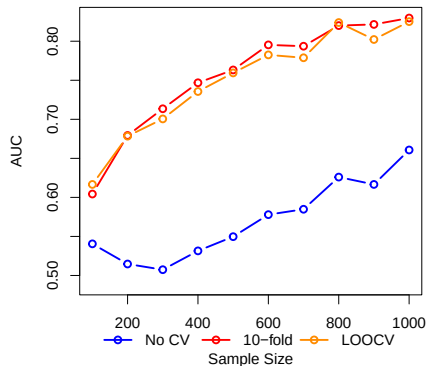


Moderate non-linear association

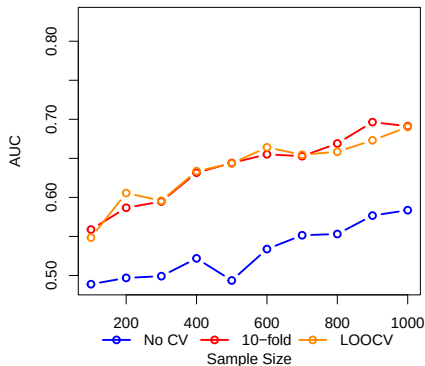


AUC of SW test P-Value

AUC of SW test P-Value, Strong Non-linearity



AUC of SW test P-Value, Moderate Non-linearity



Subsection 2

Simulation Studies and Results for Detecting Outliers

Simulation Setup

- Event times $T_{ij}^* \sim$ Weibull baseline hazard function with shape (α) parameter is 3 and scale (λ) parameter is 0.007.

$$T_{ij}^* = \{-\log(v)/[\lambda z_i \exp(x_{ij}^{(1)}) - 2x_{ij}^{(2)} + 0.5x_{ij}^{(3)}])]\}^{(1/\alpha)} \quad (5)$$

with $i = \{1, \dots, g\}$ and $j = \{1, \dots, n_i\}$, where v is simulated from uniform distribution, z_i the frailty term of all subjects in group i .

- $T_{ij} = \min(T_{ij}^*, C_i)$ is the observed survival time.
- We set up $g=10$ and n_i ($n_i = J$) from 10 to 100;
- Censoring rates: 20%, 50% and 80%.
- We generated outliers from $\exp(0.5)$, the random 10% or 10 event data were added outliers of the minimum value 2 or 4 to indicate moderate or strong deviation from the “clean” data.

Z Residual for a single dataset

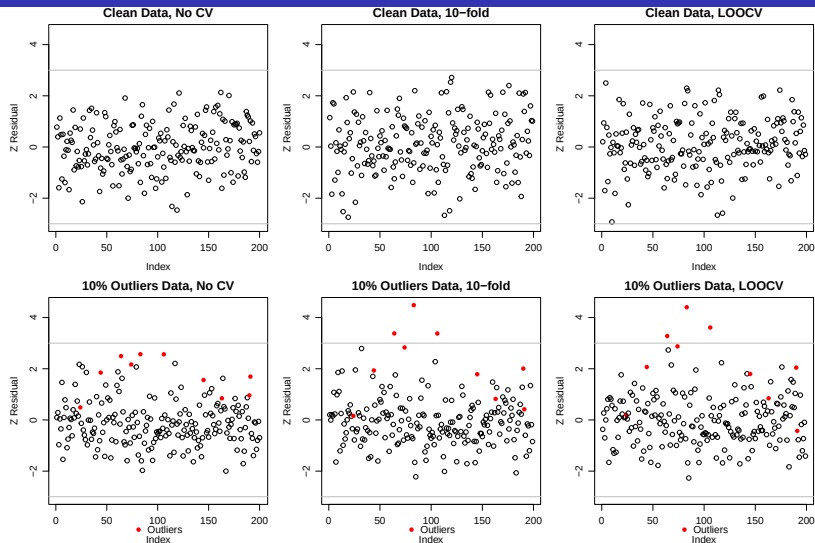
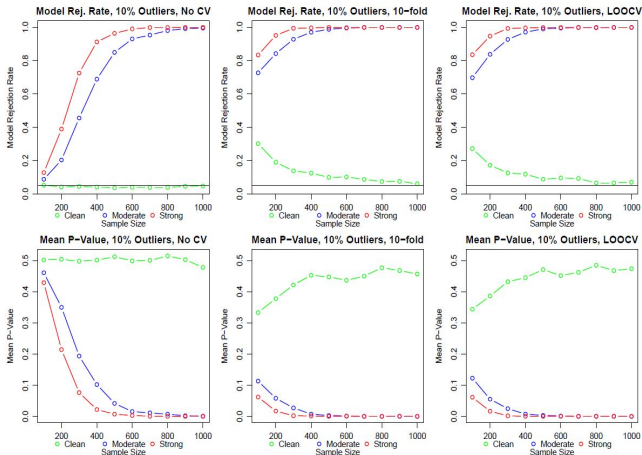
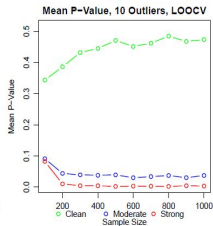
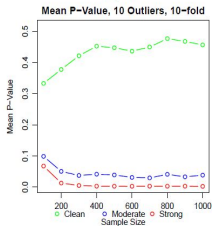
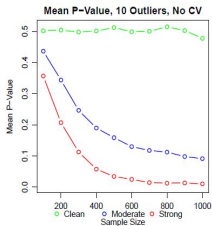
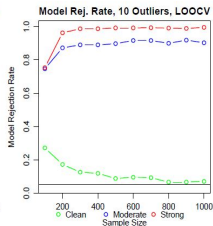
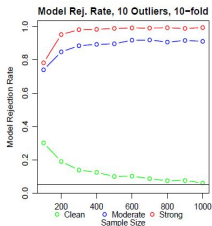
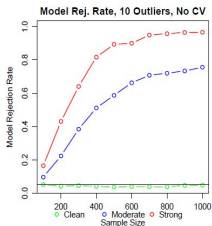


Figure 3: Sample size $n = 200$ (10 clusters of 20 observations), and censoring percentage is 50%.

10% outliers

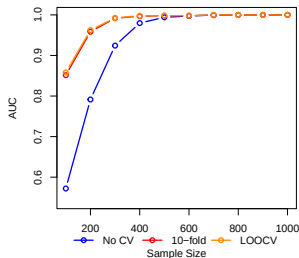


10 outliers

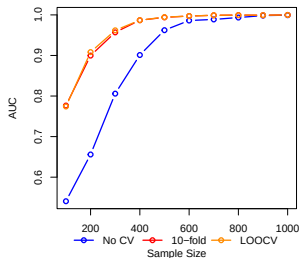


AUC of SW test P-Value

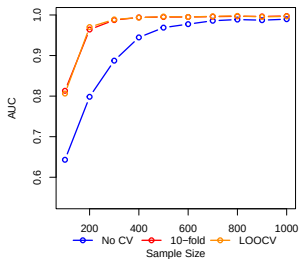
AUC of SW P-Value, 10% Strong Outliers



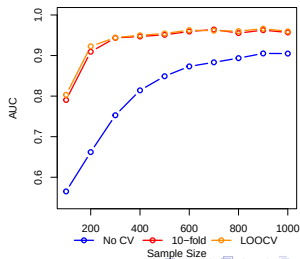
AUC of SW P-Value, 10% Moderate Outliers



AUC of SW P-Value, 10 Strong Outliers



AUC of SW P-Value, 10 Moderate Outliers



Section 4

Real Data Analysis

Data Description

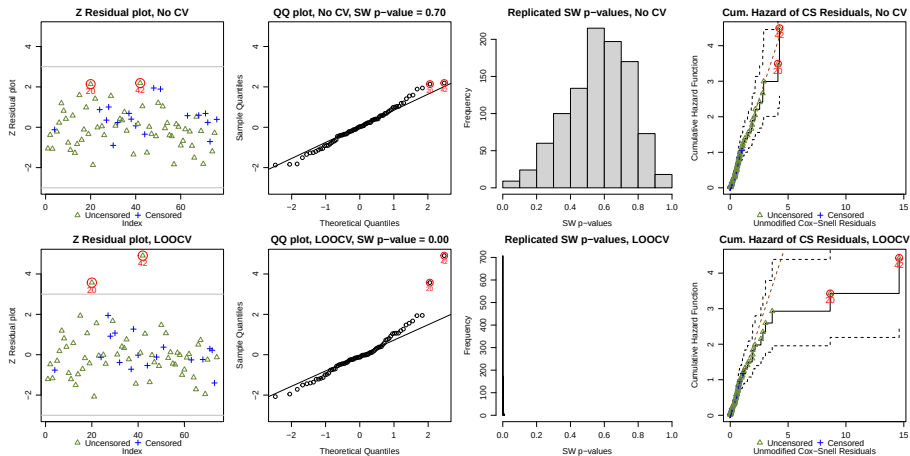
- The dataset consist of 38 kidney patients using a portable dialysis machine, and each patient is recorded times to the first and second recurrences of infection. Each survival time is the time until infection since the insertion of the catheter.
- The censoring percentage is 24%. The dataset contain 38 clusters and 2 observations in each cluster.
- We fit a shared gamma frailty model with three covariates, age, sex and disease to the recurrence failure times.
- We calculated the Z residuals and Cox Snell (CS) residuals by using the No-CV and LOOCV methods.

List of Variables

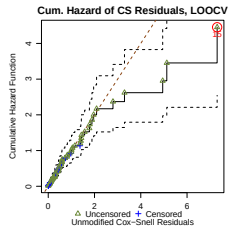
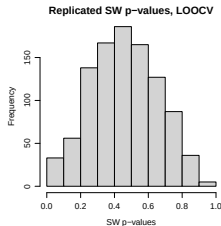
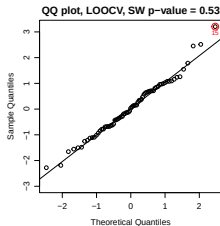
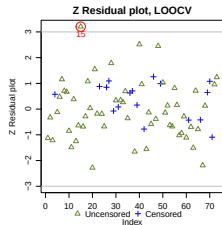
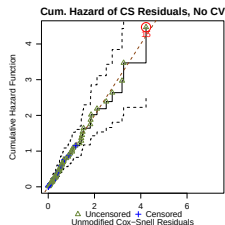
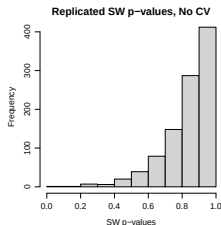
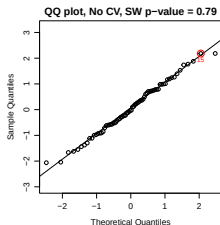
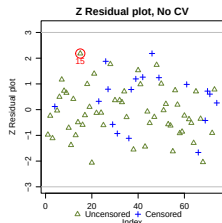
Table 1: Variable definitions for the kidney infection dataset.

Variable	Definition
<i>ID</i>	Patient number
<i>Time</i>	Recurrence time (days)
<i>Status</i>	Event indicator (1 = infection occurs; 0 = censored)
<i>Age</i>	Patient age (years)
<i>Sex</i>	Sex status (1 = male; 2 = female)
<i>Disease</i>	Disease Type (0 = GN; 1 = AN; 2 = PKD; 3 = other)

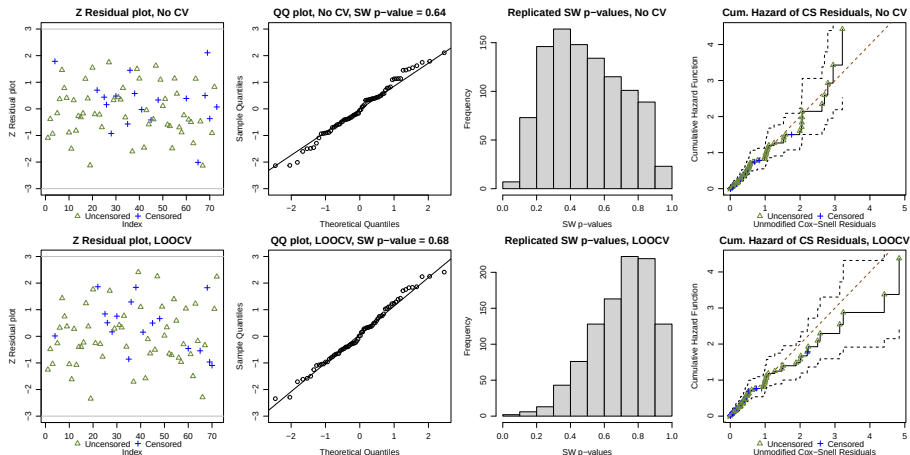
The original kidney infection dataset



The kidney infection dataset without two outliers



The kidney infection dataset without three outliers



Parameter estimates of the shared gamma frailty model

Table 2: The original kidney infection dataset.

Covariates	Estimate	SE	P-value
<i>Age</i>	0.003	0.011	0.775
<i>SexMale</i>	1.480	0.358	0.000
<i>DiseaseGN</i>	0.088	0.406	0.829
<i>DiseaseAN</i>	0.351	0.400	0.380
<i>DiseasePKD</i>	-1.430	0.631	0.023
<i>Frailty</i>			0.933

Table 3: The kidney infection dataset without two outliers.

Covariates	Estimate	SE	P-value
<i>Age</i>	0.007	0.011	0.530
<i>SexMale</i>	2.117	0.400	0.000
<i>DiseaseGN</i>	0.359	0.406	0.380
<i>DiseaseAN</i>	0.349	0.407	0.390
<i>DiseasePKD</i>	-0.797	0.638	0.210
<i>Frailty</i>			0.940

Table 4: The kidney infection dataset without three outliers.

Covariates	Estimate	SE	P-value
<i>Age</i>	0.012	0.011	0.280
<i>SexMale</i>	2.120	0.402	0.000
<i>DiseaseGN</i>	0.727	0.415	0.08
<i>DiseaseAN</i>	0.319	0.404	0.430
<i>DiseasePKD</i>	-0.802	0.636	0.210
<i>Frailty</i>			0.940

Section 5

Conclusion and Future Work

Conclusion

- Model diagnosis in survival analysis is very important but challenging.
- Our study demonstrated that Z residual is an excellent tool that can be used to diagnose shared frailty model.
- Z residuals can be regarded as an extension of randomized quantile residuals.
- Cross validation helps improving the performance of Z residuals for detecting model misspecification.

- Z residuals can be extended to diagnose the survival models with other kind of censoring mechanisms, such as interval censored data.
- The randomization in Z residuals may lower the power of Z residuals, so developing non-random test is warranted.

Acknowledgements

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